NAMAA Community at NADI 2026 shared task: Mixed-Dialect and Code-Switched Arabic ASR and Dialect Identification

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Abstract

We describe NAMAA Community's systems for three NADI 2026 shared-task subtasks: mixed-dialect automatic speech recognition (ASR), code-switched ASR, and spoken Arabic dialect identification. For mixed-dialect ASR, inference with Cohere Labs' Arabic transcription model in bfloat16 achieves 44.65% official blind-test word error rate (WER); a development-selected int8 configuration with normalisation and hypothesis selection has a previously recorded blind WER of 49%. For code-switched ASR, two-stage low-rank adaptation (LoRA) of FarukSTT achieves 20.38% validation WER and 16.98% blind WER. For dialect identification, a four-model ensemble obtains 95.00% development accuracy, while the selected three-model blind submission obtains 37.70% accuracy and 0.1674 average cost after aggregate leaderboard feedback influenced selection. We document these complete-system results and a post-hoc prediction-agreement analysis, while distinguishing them from causal component comparisons and a validated label-free selection procedure.

1 Introduction

Arabic speech processing must accommodate dialect variation, spelling variation, and multilingual speech. NADI 2026 (Sullivan et al., 2026) extends the speech-processing setting of NADI 2025 (Talafha et al., 2025). We participated in mixed-dialect Arabic ASR (Subtask 1.2), Tunisian Arabic–French–English code-switched ASR (Subtask 1.3), and spoken dialect identification (Subtask 2). The first two tasks generate transcriptions; the third assigns one of 20 Arabic dialect labels, including Modern Standard Arabic (MSA).

Our systems use established pretrained speech models: Cohere Transcribe Arabic (Cassini et al., 2026), a Tunisian adaptation of Whisper (Battikh, 2024; Radford et al., 2023), and an ensemble of speech classifiers. We report the configurations that were evaluated, the settings preserved in our records, and the limitations of the comparisons. The contribution is a system description with documented unsuccessful configurations and submission history. We do not infer that a particular post-processing operation caused the observed differences, or that prediction disagreement alone identifies an inaccurate model.

Related work. Whisper provides a multilingual encoder–decoder foundation for speech recognition (Radford et al., 2023); LoRA adapts selected weight matrices through low-rank updates (Hu et al., 2022). Conformer incorporates convolution and self-attention for speech modelling (Gulati et al., 2020). For dialect identification, ADI-20 extends the 17-dialect inventory to 20 classes and provides dataset and model resources (Elleuch et al., 2025a). ECAPA-TDNN (Emphasized Channel Attention, Propagation and Aggregation in Time Delay Neural Networks) (Desplanques et al., 2020) and its VoxLingua107 language-identification checkpoint (Valk and Alumäe, 2021) provide another pretrained starting point. We use SpeechBrain (Ravanelli et al., 2024) for this model family. The public Elyadata Whisper classifier is documented by Elleuch et al. (2025b).

2 Task, Data, and Evaluation

Mixed-dialect ASR. NAMAA-trained components used organiser-released NADI data; official blind scores came from CodaBench.[1] No task training split is provided. The public development repository, whose actual name contains `subtask2`, has 3,152 examples; the blind set has 427 clips. Our earlier records described the development set as 3,000 examples. The evaluated identifier list is unavailable, so we cannot resolve whether that count

[1]https://nadi.dlnlp.ai/2026/

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was shorthand or a local subset. Appendix A gives the repository and revision checked during this revision. The separately trained Whisper comparison used NADI 2025 Subtask 2 ASR data, described in our records as approximately 15 hours across eight dialects; the exact dataset revision remains unrecorded.

Code-switched ASR. We used `fbougares/NADI_TUN_ASR_2026`. Stage 1 used the training split. Stage 2 combined it with the publicly labelled legacy `test` split. The 731-example `validation` split was excluded from training and used for checkpoint and decoding selection. Our recorded Stage 2 pool has 25,730 source examples and 12,376 examples tagged as containing Latin-script text; duplicating the latter produces 38,106 training instances. The current public train-plus-legacy-test count is 25,735. The five-example difference, filter accounting, and precise Latin-script detection rule cannot be reconstructed from the available record.

The organisers explicitly permitted training on the labelled legacy split. The task's public participation page separately describes this split as available for experimentation and tutorials, and distinguishes it from the competition blind test.[2] An identifier/audio-overlap audit is not documented here. The 2,123-example competition blind set was used for inference.

Dialect identification. Our two trained classifiers used the approximately 200-hour ADI-20 micro release. The public classifiers were trained on ADI-20 or ADI-17. The organisers describe the 878-clip blind evaluation as out of domain and require predictions in the 20-class output space; the blind recordings need not cover every class. Training-set label counts are unavailable; Appendix A.1 gives the output labels and mappings. Development overlap with the public classifiers' training data has not been audited.

Metrics and result versions. We report WER, character error rate (CER), and accuracy as percentages; language-recognition-evaluation (LRE) average cost remains on its original scale. Subtask 1.2 scoring normalises whitespace only. Subtask 1.3 applies the organiser's transcription cleaning to references and predictions. Table 1 uses public CodaBench scores checked on 10 September 2026. These more precise and secondary scores were added during the paper revision; we do not imply that they first became available after submission. Earlier rounded results are identified separately in the appendices. Local validation and blind scores are not treated as a matched comparison of dataset difficulty.

[2] `https://www.codabench.org/competitions/17167/`

3 Systems and Results

3.1 Mixed-Dialect ASR

Model attribution and decoding. The underlying model is `CohereLabs/cohere-transcribe-arabic-07-2026`, developed by Cohere and Cohere Labs (Cassini et al., 2026). Its model card describes approximately two billion parameters, a Conformer-based encoder–decoder, Apache 2.0 licensing, and adaptation from Cohere Transcribe. NAMAA hosted the original pretrained weights and prepared a quantized variant; we did not perform additional training or fine-tuning of this model. Both precision configurations used zero-shot task inference. The model's pretraining is not a NAMAA contribution. The upstream training corpus inventory and overlap with NADI sources are not established by our records. Exact upstream and hosted checkpoint revisions remain unavailable.

We evaluated bfloat16 inference and an int8 configuration loaded with `bitsandbytes`. Both used deterministic five-beam decoding, a maximum of 440 new tokens, and the Transformers library (Wolf et al., 2020). Here *bfloat16* denotes numerical precision, not a model-size variant. The selected bfloat16 system used no orthographic post-processing or hypothesis combination.

Normalisation and hypothesis selection. The development pipeline normalised hypotheses toward development-reference spelling conventions: alef unification, alef-maqsura to yeh, taa-marbuta to haa, tatweel/diacritic removal, and hamza unification. References were left unchanged. Candidate hypotheses were combined by selecting one with minimum mean edit distance to the others. This is *medoid hypothesis selection*. Recognizer Output Voting Error Reduction (ROVER) (Fiscus, 1997) instead aligns hypotheses into a word-transition network and votes over that network; our selector only returns an existing candidate. The exact candidate inventory, distance tokenisation, and tie-breaking rule are absent from the preserved record.

Results and scope. Recorded development WERs are 56% for int8 inference and 55% for the normalisation-plus-medoid configuration. The

Subtask	Selected system	WER (%)	CER (%)	Acc. (%)	Avg. cost
1.2	Cohere, bfloat16, no post-processing	**44.65**	17.16	—	—
1.3	FarukSTT, merged Stage 2 checkpoint 200	**16.98**	6.58	—	—
2	Three-model ensemble, submission 4	—	—	**37.70**	0.1674

Table 1: Selected official blind results, retrieved during revision. Primary metrics are bold. CodaBench submission IDs are 864494, 862167, and 861912 respectively. Subtask 1.2 WER was previously rounded to 0.45 and Subtask 2 accuracy to 0.38. This table changes reporting precision, not predictions. Appendix B.2 records sources and additional metrics.

public development row for the latter gives 55.48% at higher precision. Its recorded blind WER is 49%, whereas the selected bfloat16 configuration has 44.65% official blind WER. The bfloat16 development WER and int8-only blind WER are unavailable (Table 4). Thus, these results compare complete configurations. They neither isolate normalisation, quantisation, or selection effects nor establish a matched development-to-blind ranking reversal. The records also do not establish that every configuration was fixed before any blind score was observed.

3.2 Code-Switched ASR

Starting model. FarukSTT is credited to Faruk Battikh (Battikh, 2024). The current card at `medyas/FarukSTT` identifies Whisper-large-v3 as its base and `FARUKxAUTO/tunisian-asr-cleaned` (54,156 audio–text pairs) as upstream training data. This documents the public model lineage, not a verified revision of the exact checkpoint used in our run. We retained the Whisper-large-v3 multilingual processor and tokenizer.

Adaptation. Stage 1 applied the organiser's `clean_transcription` preprocessing and trained rank-32 LoRA on query and value projections for one epoch at learning rate 5×10^{-6}. We merged the adapter into the starting model. Stage 2 added rank-32 adapters to the attention, feed-forward, and output projections, with learning rate 3×10^{-6}, using the duplicated pool described above. Both stages used bfloat16. Stage 2 used one graphics processing unit (GPU), batch size 10, and two gradient-accumulation steps; we evaluated every 200 optimiser steps. The selected checkpoint was step 200, and a later step-400 checkpoint was evaluated but not selected.

Under this batch configuration, 200 steps correspond to nominally $200 \times 10 \times 2 = 4{,}000$ sampled training instances, or about 10.5% of the 38,106-instance pool. This is a calculation from the recorded settings, not a count of unique utterances or a recovered data-loader trace. Consequently, pool construction alone does not establish how many Latin-script examples the selected checkpoint encountered.

Decoding and results. Training labels retained Arabic-language and transcription-task prefixes. Inference disabled forced decoder IDs and token-suppression constraints. The multilingual vocabulary permits tokens from several languages, but this does not establish language-specific recognition quality. Complete local validation used deterministic three-beam search, length penalty 1.0, and maximum length 256, obtaining 20.38% WER and 8.06% CER. Blind submission 862167 obtains 16.98% WER and 6.58% CER. Zero-shot FarukSTT and Stage-1-only validation scores are unavailable in the supplied record. We therefore report the adapted system's score without attributing gains to either stage or claiming demonstrated preservation of each language. Appendix B gives the recorded training and decoding settings.

3.3 Spoken Dialect Identification

Our initial development ensemble contained four classifiers: (i) our ECAPA-TDNN adapted from SpeechBrain's VoxLingua107 checkpoint; (ii) our Whisper-large-v3 LoRA classifier; (iii) `Elyadata/ADI-whisper-ADI20` (Elleuch et al., 2025b); and (iv) the 17-class `mesoj8/whisper-large-v3-arabic-dialect-30s` model (Meso, 2026). Our two classifiers obtained 80.57% and 85.53% development accuracy respectively. The public classifiers' individual development scores are unavailable. Four-way weighted soft voting, tuned by a simplex grid with step 0.1, gave 95.00% development accuracy and 0.0507 average cost.

Output alignment. The canonical order beginning MSA, BAH, TUN is explicitly specified by the organisers.[3] It was not introduced to accommodate the 17-class classifier. Public model files

[3] https://www.codabench.org/competitions/17168/

checked during revision show a different native order for Elyadata and the 17-class order for `mesoj8`. Appendix A.1 gives the required permutations. Our records describe alignment to the canonical space and zero probability for the three labels absent from `mesoj8`. This documented intent and the public label files do not establish that the submitted implementation performed every transformation correctly. Exact submitted revisions, weights, pooling/head details for our Whisper classifier, and probability conversion code have not been recovered.

Numerical checks. Our records describe degenerate validation distributions during mixed-precision loading of the public classifiers and a correction that promoted model weights to float32. We retain this as a run-specific implementation observation. It is not evidence that all current versions of those classifiers exhibit the same fault, and it does not replace an audit of label alignment.

Post-hoc prediction diagnostics. On the blind clips, the recorded Elyadata predictions assigned 63% of examples to TUN, MSA, and BAH. Top-1 agreement—the fraction of clips receiving the same predicted label—was 19–28% between Elyadata and the other classifiers, versus 37–42% among the remaining three. This establishes an unusual prediction profile relative to this ensemble, not lower accuracy by itself. Equal classifier accuracy does not imply equal pairwise agreement. In particular, `mesoj8` cannot predict the three labels prominent in the Elyadata outputs, so agreement depends on unequal label support and class marginals; the blind class prior is unknown.

The recorded four-model submission scored 31% blind accuracy. A subsequent submission without Elyadata scored 36%; another three-model variant, submission 4, scored 37.70% in the public result row. The exact weight/settings difference between these latter two submissions is unavailable, so we do not describe the change as a reproducible refinement method. Aggregate blind scores influenced selection, and submission 4 was retained as the best of ten recorded submissions (Appendix B.3). These scores do not evaluate a prespecified, single-shot, label-free selection procedure.

4 Discussion

The systems illustrate different practical requirements: spelling-sensitive transcription, multilingual transcription, and classification with heterogeneous label inventories. Their results support complete-system comparisons. The missing matched cells for Subtask 1.2 prevent identification of a development-ranking reversal; the missing FarukSTT baselines prevent attribution of the code-switched result to the adaptation stages.

For dialect identification, distribution shift is part of the task design, but the cause of this ensemble's performance difference is unresolved. Development overlap with public-model training data, incorrect output alignment, probability handling, and ordinary domain sensitivity remain possible explanations. The published native label orders make a concrete alignment audit possible; they do not confirm or refute a bug in the historical submissions. A separate labelled shifted set, with a prespecified exclusion rule and controls for label support and class marginals, would be needed to evaluate agreement-based selection.

The available error analysis is limited to the reported class concentration, agreement ranges, and aggregate unsuccessful configurations. The materials available for this revision contain no aligned ASR outputs or labelled per-example dialect predictions from which to compute language-specific errors, switch-boundary errors, per-dialect confusion, or qualitative examples. Such examples and breakdowns cannot be inferred from aggregate WER or accuracy. No significance claim is made for differences between the recorded submissions; a paired comparison would require their per-example predictions.

With three months, we would recover artifacts, run matched ASR and overlap tests, and evaluate a prespecified dialect rule on labelled shifted data.

5 Conclusion

We report 44.65% blind WER for Cohere-based mixed-dialect ASR, 16.98% WER for adapted FarukSTT code-switched ASR, and 37.70% accuracy for the selected dialect ensemble. Negative configurations and leaderboard-informed selection show that the observed differences require controlled evaluation and do not establish causal component effects or a general selection rule.

Limitations

Experiments were single runs without seed-variance analysis. The record lacks matched component ablations, language-specific error analysis,

complete preprocessing provenance, and several inference artifacts. The exact Subtask 1.2 development subset, Cohere checkpoint revisions, and public-model training overlap remain unresolved. An audio-overlap audit is not documented. The selected Stage 2 checkpoint covers only a nominal fraction of its training pool. Dialect ensemble selection used repeated aggregate blind feedback; its selected score is therefore not an independent estimate of a prespecified selection rule. The original 51% dialect-ID baseline attribution cannot be reconciled with the public organiser-account row checked during revision; Appendix B.2 preserves this discrepancy. Public model revisions cited for documentation must not be assumed to be the historical inference revisions. These gaps limit reproducibility and the interpretation of the results.

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A Data and Model Provenance

All public metadata checks in this appendix were performed on 10 September 2026. Revisions below identify the documentation inspected during revision, not necessarily the snapshots used in the original experiments.

Dataset identifiers. The Subtask 1.2 development repository is `https://huggingface.co/datasets/UBC-NLP/NADI2026_subtask2_MixedASR`, revision `e93aa9b`. Its official name indeed contains `subtask2`; the public metadata and competition instructions both specify 3,152 examples. The recorded count of 3,000 is not a verified post-filter count. The blind repository is `https://huggingface.co/datasets/UBC-NLP/NADI2026_subtask1.2_MixedASR_test` (427 clips).

For Subtask 1.3, `https://huggingface.co/datasets/fbougares/NADI_TUN_ASR_2026`, revision `4c4afb8`, lists 24,893 training, 731 validation, and 842 legacy-test examples. Table 2 distinguishes those source counts from the recorded training pool. The competition blind repository is `https://huggingface.co/datasets/UBC-NLP/NADI2026_subtask1.3_codeswitched_asr` (2,123 clips). Distinct repository/split names do not constitute an audio-overlap check.

Our Subtask 2 training source is `https://huggingface.co/datasets/UBC-NLP/NADI_2026_ADI20_micro`; the blind source is `https://huggingface.co/datasets/UBC-NLP/NADI2026_subtask2_adi_test` (878 clips). ADI-20 is documented by Elleuch et al. (2025a). Our Subtask 1.2 Whisper comparison used NADI 2025 Subtask 2 training data (Talafha et al., 2025), recorded as approximately 15 hours across eight dialects. Dataset commits, example lists, and overlap checks for that run are unavailable.

Cohere provenance. The original model citation identifies revision `0a8193c` with DOI 10.57967/hf/9549 (Cassini et al., 2026). Its public metadata identifies `CohereLabs/cohere-transcribe-03-2026` as the upstream base and approximately 2.07 billion stored parameters. The model card specifies Apache 2.0. Our recorded loading identifier was `NAMAA-Space/cohere-transcribe-arabic-07-2026`. NAMAA hosted the original weights and prepared an int8 variant without additional training or fine-tuning. Exact source and hosted commit identifiers and the quantization conversion configuration remain unavailable. An unauthenticated check did not expose the hosted repository's metadata. We do not claim authorship of the pretrained model or verified absence of pretraining overlap.

Subtask 1.3 count	Examples
Current public training split	24,893
Current public legacy test split	842
Current public combined source	25,735
Recorded Stage 2 source pool	25,730
Unreconciled difference	5
Recorded Latin-script subset	12,376
Recorded duplicated training pool	38,106
Held-out validation split	731

Table 2: Public source counts versus recorded pool counts. Filtering of overlong audio/labels was reported, but neither the affected identifiers nor pre-/post-filter accounting are available in the supplied record. We do not assert that filtering explains the five-example difference.

FarukSTT and public dialect classifiers. FarukSTT documentation was checked at revision `a0898fa` of `medyas/FarukSTT`. Its card credits Faruk Battikh, identifies Whisper-large-v3 and the 54,156-pair `FARUKxAUTO/tunisian-asr-cleaned` corpus, and requests a 2024 model citation. The revision used in our adaptation is not available in the supplied record.

Elyadata documentation was checked at revision `04f1392` of `Elyadata/ADI-whisper-ADI20`. The public configuration specifies a Whisper-large-v3 encoder, attention pooling over 1,280-dimensional features, a bias-free $1280 \rightarrow 20$ linear head, and log-softmax output. The `mesoj8` card was checked at revision `65e8bb9`; it describes encoder attention pooling followed by layer normalisation, dropout, and $1280 \rightarrow 512 \rightarrow 256 \rightarrow 17$ linear layers with Gaussian Error Linear Unit (GELU) activations. These are public architecture descriptions, not a reconstruction of the submitted loader.

Label	Canonical	Elyadata	mesoj8
MSA	0	18	—
BAH	1	17	—
TUN	2	19	—
ALG	3	0	0
EGY	4	1	1
IRA	5	2	2
JOR	6	3	3
KSA	7	4	4
KUW	8	5	5
LEB	9	6	6
LIB	10	7	7
MAU	11	8	8
MOR	12	9	9
OMA	13	10	10
PAL	14	11	11
QAT	15	12	12
SUD	16	13	13
SYR	17	14	14
UAE	18	15	15
YEM	19	16	16

Table 3: Zero-based native indices from public documentation and their required canonical destinations. A dash denotes an unsupported label, not an unknown probability. The submitted mapping has not been independently verified.

A.1 Native Label Orders and Required Alignment

The organisers' canonical order is:

> MSA, BAH, TUN, ALG, EGY, IRA, JOR, KSA, KUW, LEB, LIB, MAU, MOR, OMA, PAL, QAT, SUD, SYR, UAE, YEM.

The public Elyadata `dialect_encoder.txt` instead orders the 17 labels ALG through YEM as above, followed by BAH, MSA, TUN. The `mesoj8` card orders its 17 labels ALG through YEM. Table 3 records the resulting mapping. For a native probability vector p, canonical alignment requires $q_{\pi(i)} = p_i$. Missing `mesoj8` classes receive probability zero. Zero-filling raw logits before softmax would give those classes nonzero mass and would not implement the stated procedure.

Elyadata's public configuration returns log probabilities. Converting these to probabilities requires exponentiation (or an equivalent normalisation of the log scores) before weighted soft voting. The public files establish the required representation; the historical score-conversion code remains unavailable. Neither this specification nor a correct class list verifies the submitted implementation. Development predictions and the exact loader are required for that audit.

B Implementation and Detailed Results

B.1 Recorded Hyperparameters

Mixed-dialect ASR. The bfloat16 model used `torch.bfloat16`, scaled dot-product attention (`sdpa`), and `device_map="cuda:0"`. The int8 configuration used `bitsandbytes`. Both used `num_beams=5`, `max_new_tokens=440`, and `do_sample=False`. Our records relate the token limit to reference lengths, but the calculation and implementation versions are not available in the supplied record. The Whisper-large-v3 comparison used rank-32 LoRA, scaling $\alpha = 64$, dropout 0.05, and targets `q_proj`, `k_proj`, `v_proj`, `out_proj`, `fc1`, and `fc2`; AdamW learning rate was 10^{-4} for three epochs. The experiments used an NVIDIA A100 40 GB or Google Colab T4/A100 instances; wall-clock training times were not retained.

Code-switched ASR. Both LoRA stages used bfloat16. Stage 1: official training split, one epoch, rank 32 on `q_proj` and `v_proj`, learning rate 5×10^{-6}, then adapter merging. The cleaning function removed designated punctuation, diacritics, tatweel, and unclear-speech markers, handled separators, normalised whitespace, and lowercased Latin text.

Stage 2: merged Stage 1 initialisation; rank 32, $\alpha = 64$, dropout 0; targets `q_proj`, `k_proj`, `v_proj`, `out_proj`, `fc1`, `fc2`, and `proj_out`; learning rate 3×10^{-6}; one GPU, batch 10, accumulation 2. Audio was resampled to 16 kHz. Overlong audio/labels were filtered, but thresholds and excluded identifiers are not available in the supplied record. Time-mask SpecAugment used probability 0.02 and length 10. Evaluation occurred every 200 optimiser steps. Step 200 was selected and merged; a recorded step-400 evaluation establishes that training proceeded beyond the selected checkpoint, but the final stopping step is unavailable.

Full local validation used three beams, length penalty 1.0, maximum length 256, deterministic generation, and disabled forced decoder IDs and token suppression. The retained evidence does not establish byte-identical local-validation and official-scoring normalisation. Raw blind transcriptions were submitted to the organiser's scorer. The duplicated Latin-script category is an operational label in the records; it is not a verified count of French or English examples, and may include other Latin-script material.

Subtask 1.2 configuration	Dev WER (%)	Blind WER (%)	Evidence
Whisper-large-v3-turbo, zero-shot	63.47	—	Public baseline row
Whisper-large-v3, our LoRA comparison	∼62	—	Approximate record
Cohere int8, five beams	56	—	Rounded record
Cohere int8, normalisation + medoid	55.48	49	Public dev; rounded blind
Cohere bfloat16, no post-processing	—	**44.65**	Public selected row

Table 4: Available Subtask 1.2 comparisons. Dashes denote unavailable scores, not zero. The approximate Whisper-LoRA result is retained because an exact evaluation record was not available. Public dev values come from submissions 809028 and 854784; no evaluated identifier list is available for local development runs.

Subtask 1.3 evaluation	System / decoding	WER (%)	CER (%)
Local validation, training-time	Stage 2 step 200, greedy	21.12	8.35
Local validation, training-time	Stage 2 step 400, greedy	21.34	8.45
Local validation, full split	Merged step 200, three beams	**20.38**	**8.06**
Official blind	Merged step 200, submission 862167	**16.98**	6.58
Official blind, hamza-normalised	Same submission, secondary scoring	16.68	6.46

Table 5: Code-switched ASR results. The full local validation split contains 731 examples and the blind set 2,123. Training-time greedy scores and merged-model beam scores differ in decoding. Zero-shot and Stage-1-only validation scores are unavailable.

Dialect identification. Our ECAPA model started from `speechbrain/lang-id-voxlingua107-ecapa`, replacing the head with a $192 \rightarrow 20$ linear layer. Records specify 25-ms frames and 10-ms shifts; a 9,000-step warm-up from approximately 3.3×10^{-5} to 10^{-4}, a 15,000-step hold, then unfreezing, a second warm-up, and decay to 90,000 steps. Exact schedule boundaries beyond these settings are not available in the supplied record.

Our Whisper model used rank-32 LoRA, $\alpha = 64$, dropout 0.05, query/key/value/output and feed-forward projections, AdamW peak learning rate 10^{-4}, a cosine schedule, batch 8 with accumulation 4, and 90,000 steps. Records describe two frequency and two time masks for SpecAugment and adapters in both encoder and decoder, but do not specify which representation was pooled into the 20-way classifier or its head. These details cannot be reconstructed from the base Whisper architecture alone.

The four-model ensemble used development-accuracy optimisation on a simplex grid with step 0.1. Exact weights for the development ensemble and all ten blind submissions, probability transformations, and submission 3-to-4 changes are unavailable. Our recorded checkpoint names are `FatimahEmadEldin/nadi26-ecapa-v1-ckpts` and `FatimahEmadEldin/nadi26-whisper-ckpts`. Unauthenticated metadata checks did not confirm accessible releases. We therefore report these as recorded identifiers, not verified public or immutable artifact releases.

B.2 Detailed Scores and Version Notes

The public leaderboards are available through the competition pages for Subtask 1.2, Subtask 1.3, and Subtask 2. Their public result rows checked during revision report:

- Subtask 1.2, submission 864494: WER 0.446541 and CER 0.171645, reported here as 44.65% and 17.16%.
- Subtask 1.3, submission 862167: WER 16.98%, CER 6.58%; with the separate hamza-normalised scorer, WER 16.68% and CER 6.46%.
- Subtask 2, submission 861912: accuracy 0.3769931663 and average cost 0.1673619627, reported here as 37.70% and 0.1674.

These secondary results were incorporated during revision and are not new evaluations run by our team. The higher-precision four-way development row (submission 854884) has accuracy 0.9500277624 and cost 0.0507493600. Its rounded accuracy is retained as 95.00%.

The Subtask 2 development baseline row identified in the organiser's instructions (`prsull`, submission 802896) has accuracy 0.8118637794, or 81.19%. The prose on the same competition page says approximately 81.39%; the latter matches the

#	Recorded date/time	Recorded configuration	Acc. (%)	Status
1	2026-07-24 23:28	Four-way, initial weights	31	Historical rounded score
2	2026-07-24 23:43	Four-way, alternate weighting	15	Historical rounded score
3	2026-07-24 23:58	Three-way, Elyadata removed	36	Historical rounded score
4	2026-07-25 00:16	**Three-way, alternate settings**	**37.70**	**Selected; public row**
5	2026-07-25 00:16	Sibling variant of submission 4	32	Historical rounded score
6	2026-07-27 19:17	Development-accuracy-selected configuration	28	Historical rounded score
7	2026-07-27 20:04	Further ensemble tuning	15	Historical rounded score
8	2026-07-27 20:44	Further ensemble tuning	27	Historical rounded score
9	2026-07-28 02:41	Further ensemble tuning	22	Historical rounded score
10	2026-07-28 03:56	Further ensemble tuning	28	Historical rounded score

Table 6: Complete recorded dialect-ID submission history, spanning five recorded calendar dates. Timestamp timezone is unverified; the selected public row gives 24 July 2026, 21:16:11 UTC. Submission 4 was previously rounded to 38%. Submission 6 was recorded as `best_id_accuracy_submission.zip`. Exact weights and the changes between variants are unavailable.

Dialect-ID system	Dev acc. (%)
Public organiser row (`prsull`)	81.19
Our ECAPA-TDNN	80.57
Our Whisper-large-v3 (LoRA)	85.53
Elyadata ADI-Whisper	—
`mesoj8` (17 classes)	—
Four-way development ensemble	**95.00**

Table 7: Dialect-ID development results. Dashes indicate missing individual evaluations on this split. The public classifiers' own benchmark scores are not substitutes for these missing cells.

previously reported 0.8139. We now distinguish the numeric row from that approximate prose. The public blind row for `prsull`, submission 856127, has accuracy 28.70% and cost 0.2002. Our earlier 51% blind-baseline attribution has no matching organiser-account row in this snapshot. We retain that discrepancy here and do not use the 51% figure to claim a baseline comparison.

The public Subtask 1.3 development submission 854889 has 21.21% WER and 8.44% CER, whereas our local merged-checkpoint evaluation records 20.38% and 8.06%. The public row is not substituted for the local result because a matching configuration/scoring record has not been recovered.

B.3 Blind Submission History

Table 6 preserves all ten entries in our recorded dialect-ID submission history. Aggregate feedback was available after submissions and influenced later configurations and final selection. No single classifier was evaluated alone on the blind set in this recorded history. The selected score must be interpreted in that selection context.

The recorded timestamp for submission 4 is 25 July at 00:16, while the matching public result row (861912) is timestamped 24 July at 21:16:11 UTC. The three-hour difference is consistent with a timezone conversion, but the time basis for the other entries has not been verified. We therefore no longer label the recorded timestamps as UTC. Entries 4 and 5 retain their identical recorded minute; the underlying submission IDs for the other entries are needed to check them. Exact ensemble weights and settings are not recoverable from aggregate scores or timestamps.

Submission packaging. Subtask 1.2 used a UTF-8 `predictions.txt`, one transcription per identifier in recorded ID-sorted order, at the ZIP root. Subtask 1.3 used `prediction_test.csv` with columns `file_name,transcription`, one row per blind identifier. Subtask 2 used `predictions.tsv` and `logits.tsv` at the ZIP root; the latter had 20-class columns recorded as softmax probabilities. The submitted files are unavailable, so representation and order cannot be verified.